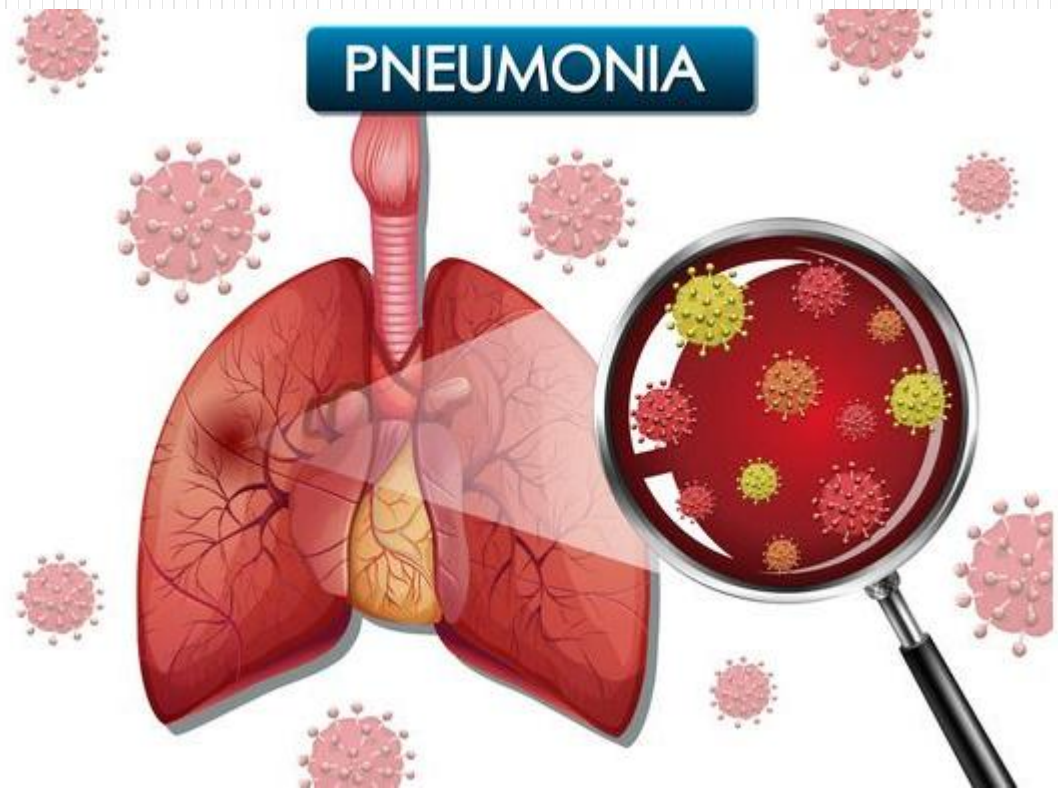


Management of patient with pneumonia and TB

Dr . Bardes Shaban





الحديث الجانبي يشتت الانتباه



من فضلك اغلق الهاتف

OUTLINES OF PNEUMONIA

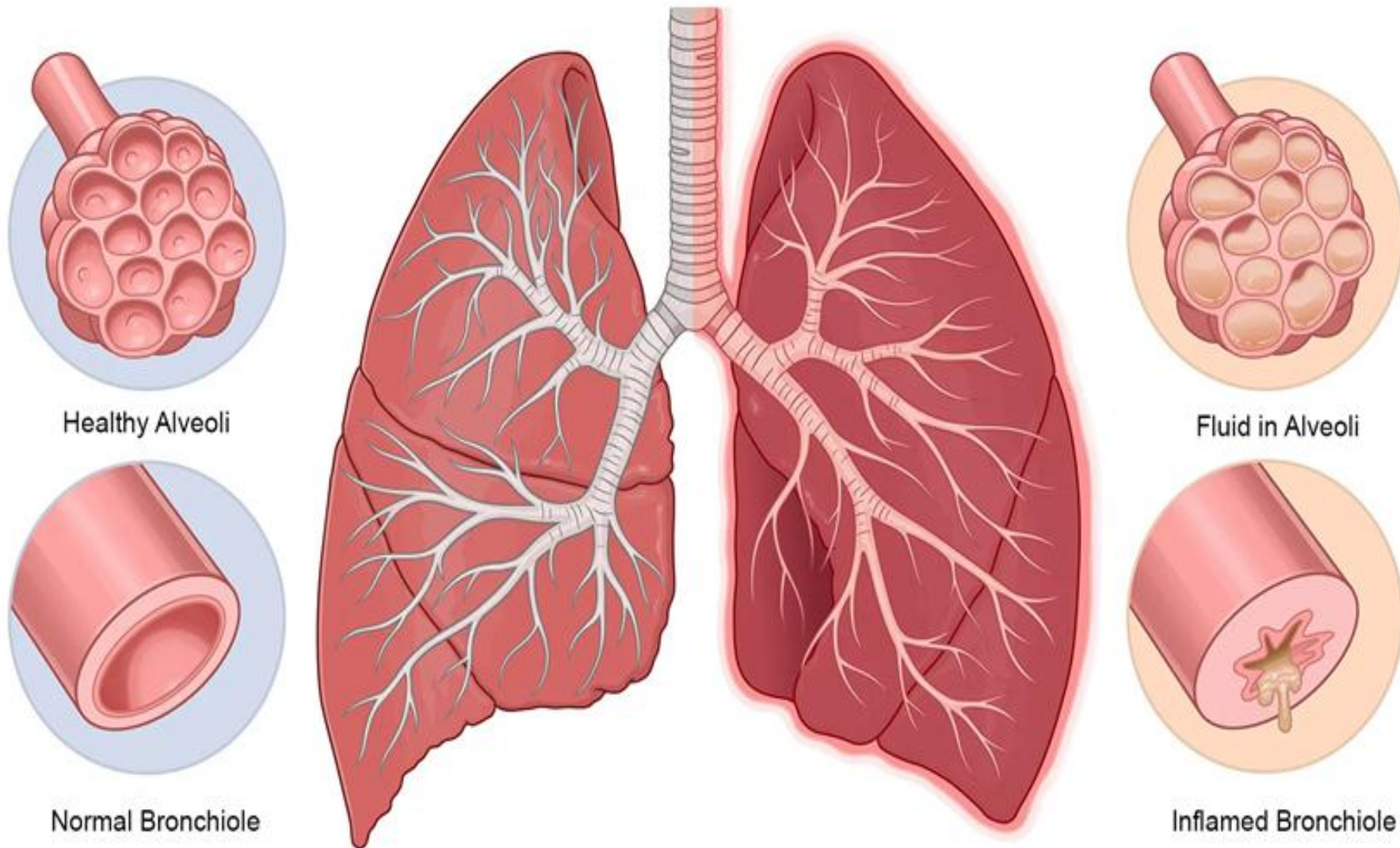
- ◉ Definition
- ◉ Classification ,Pathophysiology & risk factors
- ◉ Clinical manifestations
- ◉ Assessment and diagnoses
- ◉ Management
- ◉ Complications

◎ Definition:

Inflammation of the lung parenchyma caused by various microorganisms, including bacteria, mycobacteria, fungi, and viruses

Healthy Lung

Pneumonia



Classifications

**Community-Acquired
Pneumonia**

**Hospital-Acquired
Pneumonia**

**Pneumonia in the
Immunocompromised
Host**

Aspiration pneumonia

1- Community-Acquired Pneumonia

❖ Occurs either in the **community** setting or within the **first 48** hours after **hospitalization**.

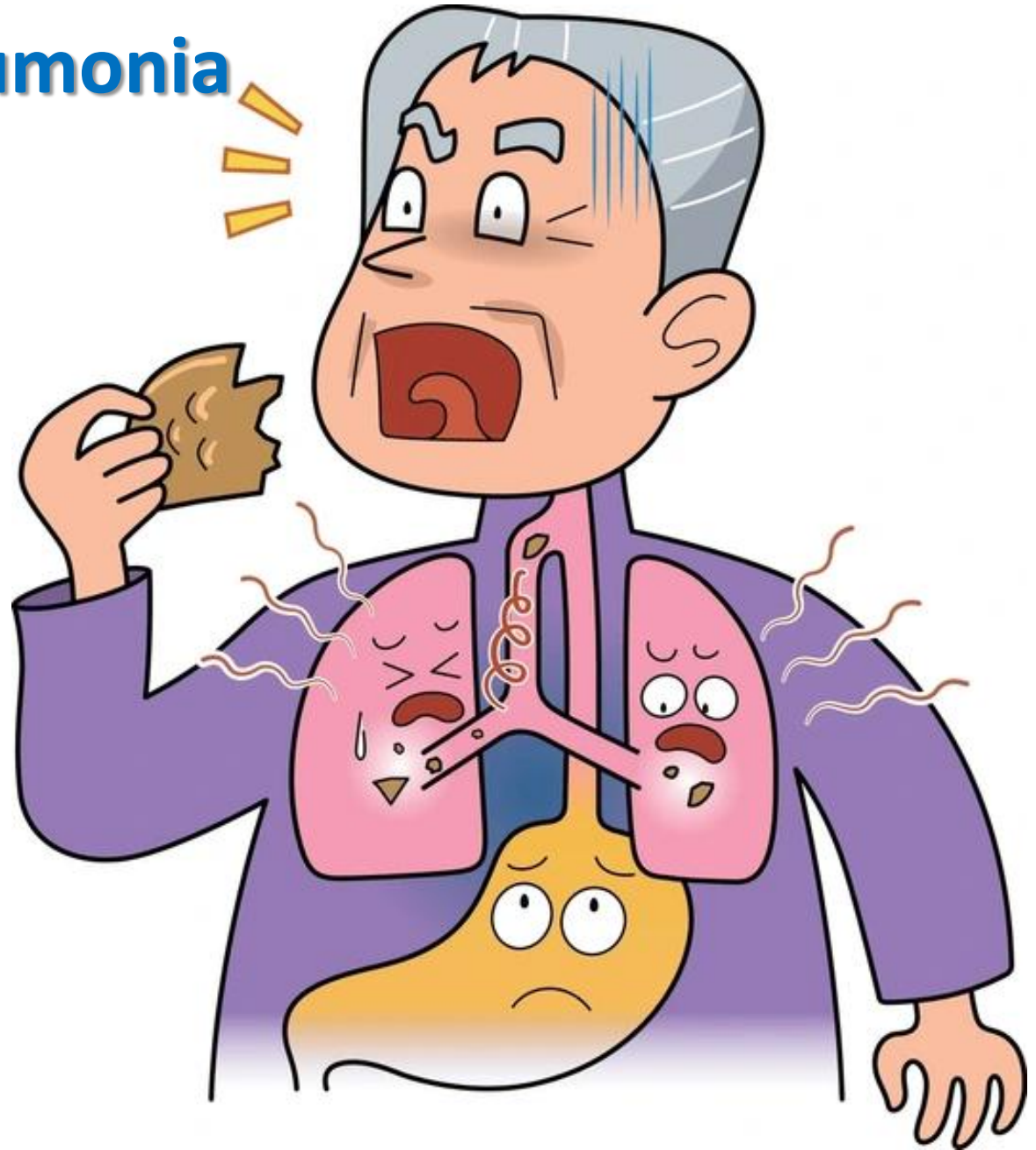
2- Hospital-Acquired Pneumonia

- ❖ Nosocomial pneumonia
- ❖ The onset of pneumonia symptoms **more** than **48 hours** after admission
- ❖ **Ventilator** associated pneumonia (endotracheal intubation and mechanical ventilation).

3- Aspiration pneumonia

- ❖ Aspiration pneumonia is an **infection** of the lung tissue ([parenchyma](#)) that develops after **foreign material** from the mouth or stomach is **inhaled** into the lungs
- ❖ It occurs when the material is **not swallowed** properly and "goes down the **wrong pipe**".
- ❖ Aspiration pneumonia may occur in the **community** or **hospital** setting.

Aspiration pneumonia



Causes / risk factors

- Conditions that produce mucus or bronchial obstruction and interfere with normal lung drainage (e.g., cancer, cigarette smoking, chronic obstructive pulmonary disease)
- Immunosuppressed patients and those with a low neutrophil count
- Prolonged immobility
- Advanced age

Causes / risk factors

- Depressed cough reflex
- Nothing-by-mouth (NPO) status; placement of nasogastric, oro gastric, or endotracheal tube
- Supine positioning
- Alcohol intoxication
- General anesthetic, sedative, or opioid

Pathophysiology

An inflammatory reaction occur in the alveoli, producing **exudate** that **interferes** with the diffusion of **oxygen** and carbon dioxide



lung are **not** adequately **ventilated** because of **secretions** that cause partial **occlusion** of the bronchi or alveoli



Bronchospasm and hypoventilation



Persistent Cough



Chest Pain



Rapid Breathing



SYMPTOMS OF PNEUMONIA

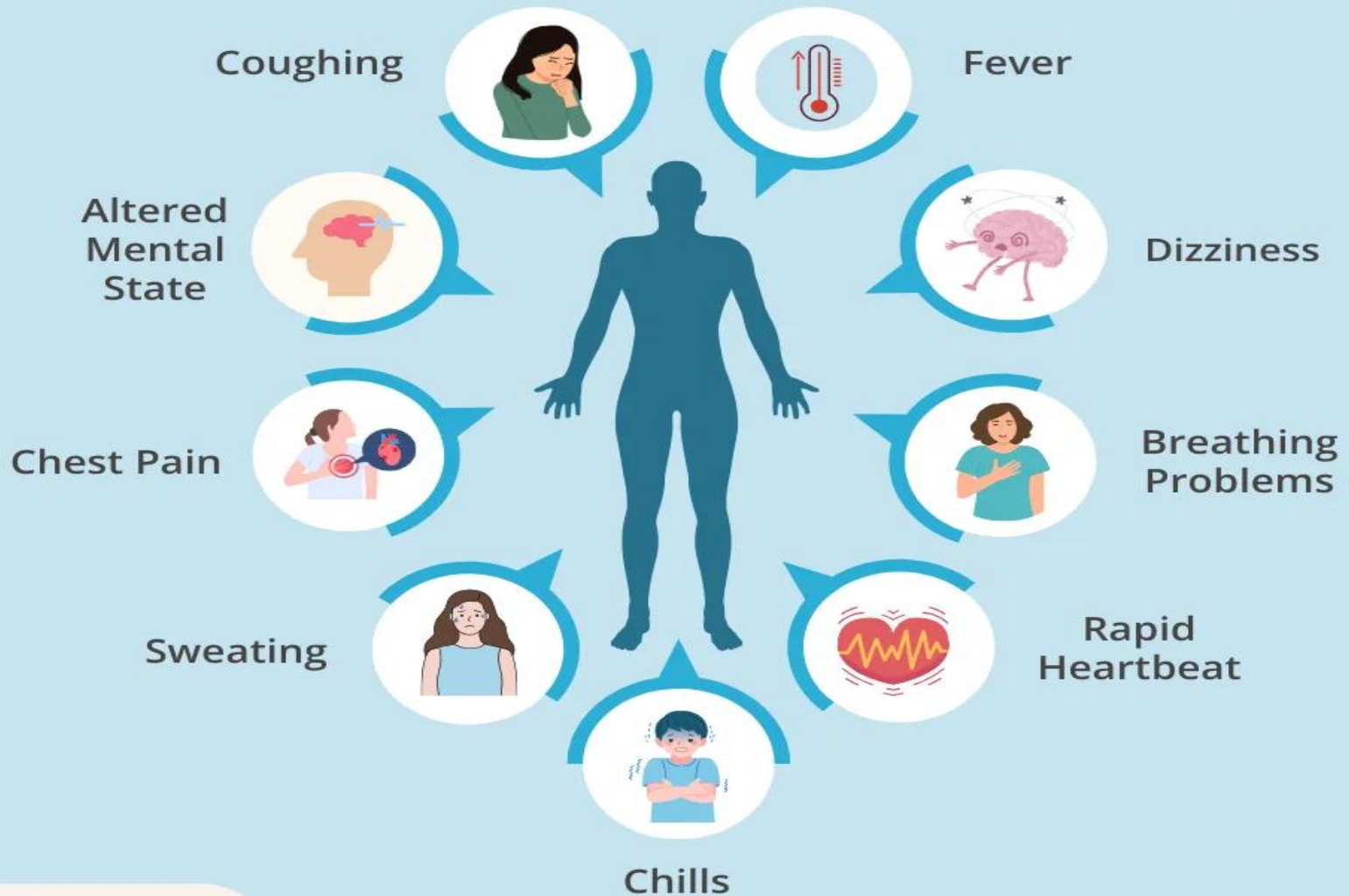


High Fever & Sweating



Mental Changes

How Pneumonia Affects the Body



Assessment & Diagnostic Findings



Physical examination:

- Presence of symptoms.

Diagnostic studies :

- Chest X-ray
- Sputum examination
- Blood culture



Prevention

Vaccination against pneumococcal infection for:

- People 65 years of age or older
- Immunocompromising people
- People with functional or anatomic asplenia
- People living in environments or social settings in which the risk of disease is high.
- Immunocompetent people who are at increased risk for illness because of chronic illness

- Promote coughing and expectoration of Secretions
- Initiate suctioning and chest physiotherapy .
- Encourage smoking cessation.
- Reposition frequently to promote lung expansion , elevate head of bed at least 30 degrees.
- Promote frequent oral hygiene.

Treatment

Medical

- **Antibiotics** are indicated with a viral respiratory infection only if bronchitis, or rhino sinusitis is present.
- **Hydration** is a necessary part of therapy because fever and tachypnea may result in insensible fluid losses

- **Antipyretics** to treat headaches and fever
- Warm, moist **inhalations** are helpful in relieving bronchial irritation
- **Antihistamines** to reduced sneezing and rhinorrhea
- **Oxygen** therapy if hypoxemia develops



Nursing process for patients with pneumonia

Assessment

- **Clinical manifestations** of pneumonia: fatigue, tachypnea, use of accessory muscles for breathing, coughing, and purulent sputum.
- Assess **elderly** patients for unusual behavior, altered mental status, dehydration, excessive fatigue.

Nursing Diagnoses

- Ineffective airway clearance related to secretions
- Activity intolerance related to impaired respiratory function
- Risk for deficient fluid volume related to fever and a rapid respiratory rate.
- Deficient knowledge about the treatment regimen and preventive health measures.

Planning and goals



Improved airway patency



Conserve energy



Maintenance of proper fluid
volume

Understanding of the treatment and
preventive measures

Nursing interventions

Improving airway patency

- Deep breathing & coughing exercises , using incentive spirometer.
- Correct positioning for lung expansion.
- Chest physiotherapy (percussion , vibration and postural drainage).
- Humidification may be used to loosen secretions and improve ventilation.

Conserving energy

- Promote rest
- Moderate activity

Promoting fluid intake

- Encourage increased fluid intake (at least 2 L/day), unless contraindicated as in heart failure

Promoting patients' knowledge

- Inform the patient and family about the cause, symptoms, management of pneumonia and the need for follow up.
- If the patient is hospitalized, is instructed about the importance of adhering management during and after the hospital stay
- Explanations should be given simply and in language that the patient can understand.
- Written instructions should be provided.

Evaluation

- Improved airway patency
- Rests and conserves energy
- Maintains adequate hydration
- Exhibits no complications

Complications

respiratory failure

Congestive heart failure

Atelectasis

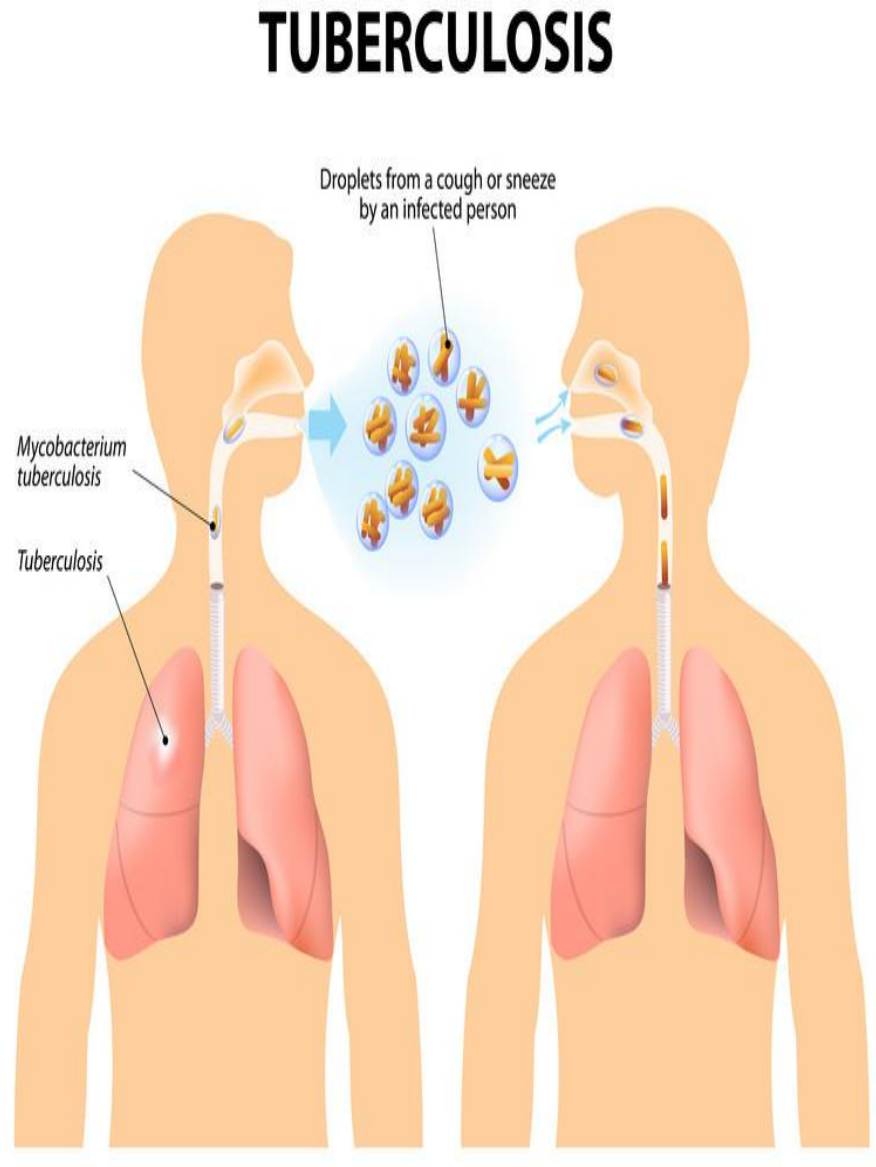


Summary



- **List 3 types of pneumonia**
- **List 4 clinical manifestations of pneumonia**
- **List 4 complications of pneumonia.**

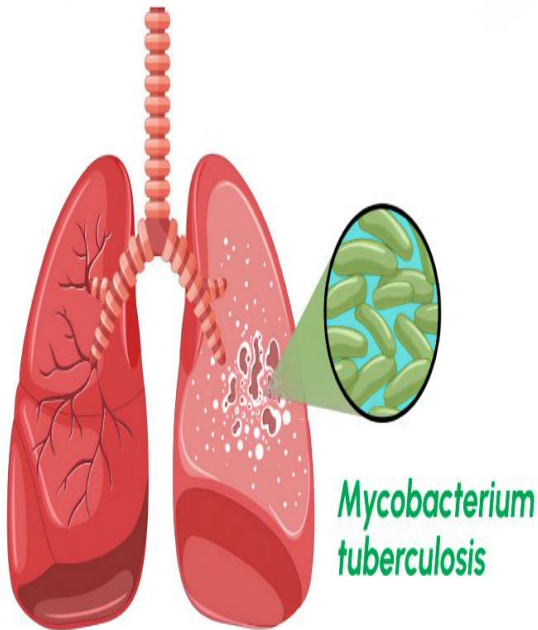
Tuberculosis (TB)



Outlines

- Definition
- Causes
- Clinical manifestations
- Assessment and diagnoses
- Management
- Complications

What is Tuberculosis (TB) ?



Definition

- **Tuberculosis (TB):**
is an infectious disease caused mainly by **Mycobacterium tuberculosis**. It primarily affects the lung parenchyma, but it can spread to almost any part of the body as , kidneys, liver , brain and

Causative Agent

Caused
by **Mycobacterium
tuberculosis.**

Transmitted
via **airborne
droplets** when
infected person
coughs or sneezes.

Stages of TB

**Early infection
primary TB**

Latent TB infection

Secondary TB

TUBERCULOSIS SIGNS AND SYMPTOMS



How to diagnose

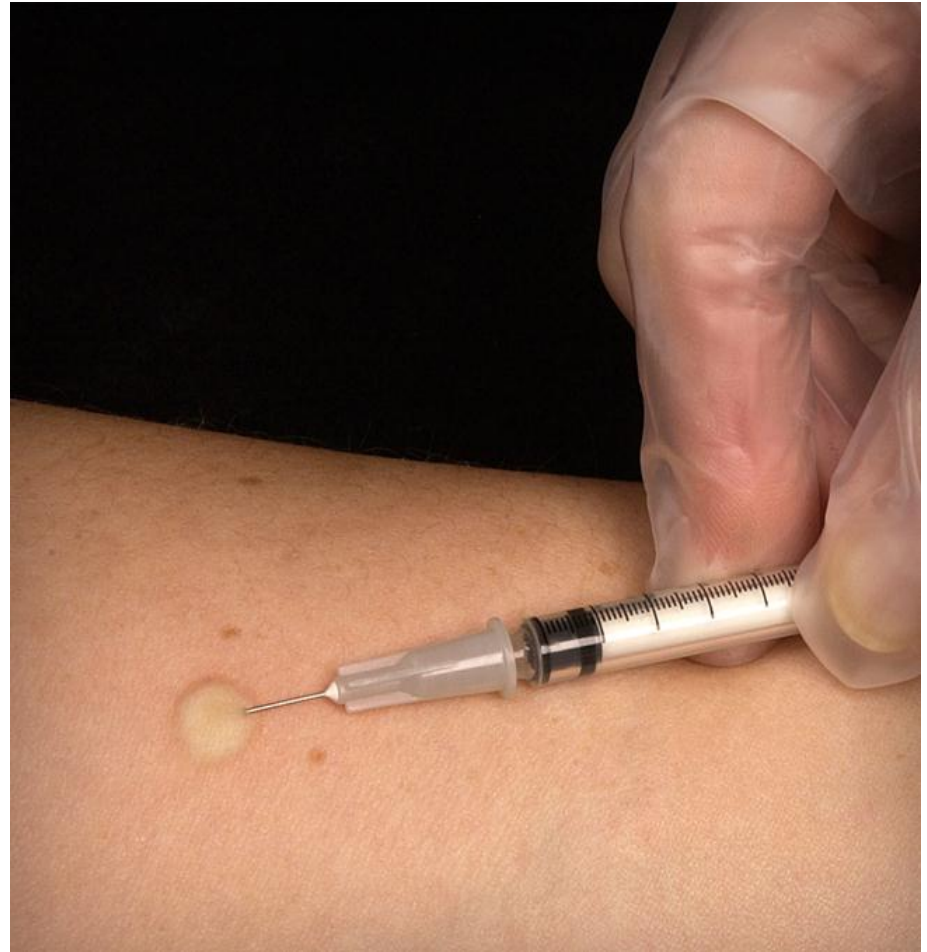
- o History taking
- o Physical examination
- o Tuberculin skin test.
- o Chest X-ray and CT scan for confirmation.
- o Bronchoscopy.
- o Sputum culture

TB diagnosis should be considered for any Patient who has :

- Persistent cough lasting longer than 3 weeks, chest pain, weakness, weight loss, anorexia, hemoptysis, dyspnea, fever, night sweats, or chills

TB skin test (Mantoux test)

- A dose of 0.1 ml of purified protein derivative (PPD) is injected **intradermally** into the forearm.
- **After 48 to 72** hours, the injection site is examined for induration.
- If the induration measures 10 mm or more, the test is considered positive.



PREVENTIONS



STAY AWAY FROM COUGHING PEOPLE



CLEAN YOUR HOME



WASH YOUR HANDS



USE PERSONAL DISHES

BCG Vaccine

Nursing management

- **Administer antitubercular medications** to treat and prevent transmission: (isoniazid, rifampin, pyrazinamide, ethambutol, streptomycin)
- **Respiratory isolation** for in-hospital care: the bacteria are spread by droplet.
- **Increase protein**, carbohydrates, and vitamin C diet for patients
- **Supplementation with vitamin D** activates the immune system.
- Encourage patient to **stop smoking**.

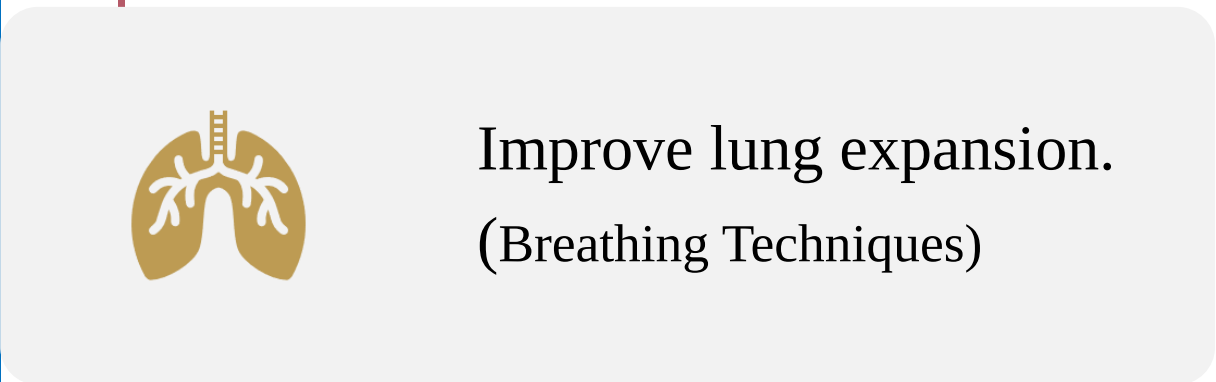
- **Isolate** patient in properly ventilated room and maintain. Obtain three early-morning sputum samples.
- Provide **sputum samples** every 4 weeks to monitor the effectiveness of the medication.
- Wear **personal protective equipment** when obtaining specimens
- **Monitor** respiration for rate, effort, use of accessory muscles, and skin color changes.
- Increase **fluid intake** to help liquefy any secretions. Record fluid intake and output.
- Perform **chest physiotherapy** to patient several times per day. Encourage Coughing and **deep breathing** exercises.

Physiotherapy



clearance(postural
&&&
physiotherapy).

drainage
chest



Improve lung expansion. (Breathing Techniques)

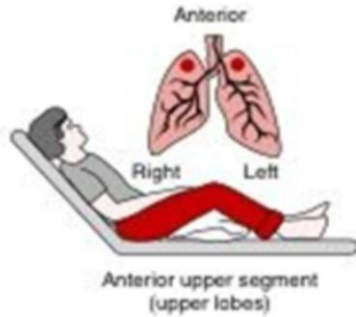


Prevent muscle wasting

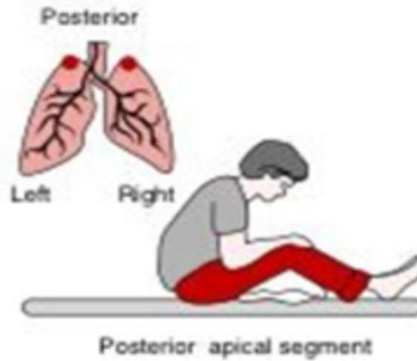


Postural drainage, **percussion**, and **vibration**.
Breathing exercises and energy conservation.

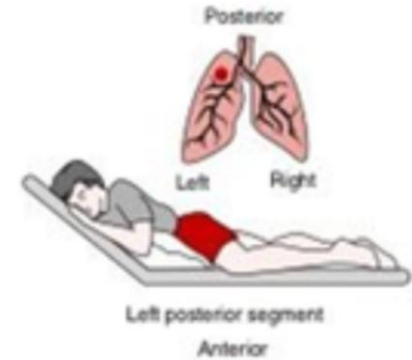
What position should I use?



Sit semi-upright (about 45 degrees) and back flat to the bed/chair, place pillows under your knees to bend your legs slightly. How long:



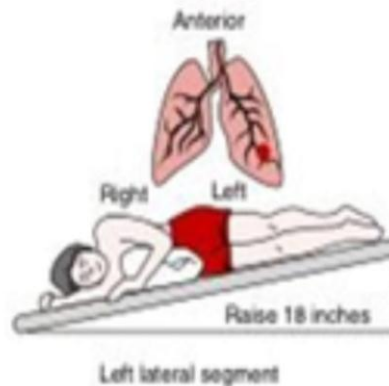
Sit upright with a pillow below your knees to have a slight bend in them. Lean forward through your forearms across your thighs. How long:



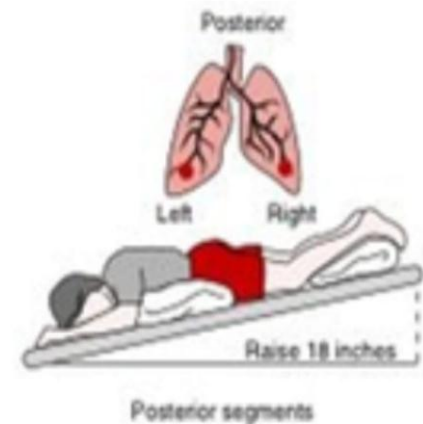
Lay flat onto your front with pillows to support up to around 45 degrees. Place a small pillow under the left side of your chest so you have a slight lean to the right. How long:



Lay flat onto your back with a pillow under your right side to lean to your left slightly. Place a small pillow under your waist and hips to promote a slight downwards lean. How long:



Lay onto your right side with a pillow supporting your left arm. Place a small pillow under your waist and hips to promote a slight downwards lean. How long:



Lay flat onto your front with a pillow under your stomach for comfort. Place a small pillow under your waist and hips to promote a slight downwards lean.



Medical Management

4 drugs for 2 months

RIPE regimen

Rifampicin.

Isoniazid.

Pyrazinamide.

Ethambutol.

2 drugs for next 4:6 months

Rifampicin.

Isoniazid

Several antibiotics such as (Capreomycin, Kanamycin, and Amikacin) are given every day for up to two years

Complications

- Pleurisy, pericarditis, peritonitis.
- Meningitis.
- Bone and joint infections,
- Genitourinary or gastrointestinal infection.
- Bronchiectasis.





- **Define TB.**
- **List 4 clinical manifestations of TB.**
- **List 4 complications of TB.**
- **Discuss Physiotherapy role in TB**

thank
you